

Yiyang Huang

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EDUCATION

Northeastern University, Boston, MA <i>Ph.D. in Computer Engineering, advised by Prof. Yun Raymond Fu</i>	Sep 2024 – Aug 2029 (Expected)
Xidian University, Xi'an, China <i>M.S. in Computer Science, advised by Prof. Xuefeng Liang</i>	Sep 2021 – Jun 2024
Xidian University, Xi'an, China <i>B.Eng. in Intelligence Science and Technology</i>	Sep 2017 – Jun 2021

RESEARCH INTERESTS

Multimodal LLMs and Agents; Long-Video Understanding; Hallucination Detection and Mitigation; Agentic Reasoning, Reflection, and Tool Use; Self-Improving Multimodal Agents

RESEARCH EXPERIENCE

Northeastern University | SMILE Lab | Research Assistant | Boston, MA
Supervisor: Prof. [Yun Raymond Fu](#) Sep 2024 – present

[\[Preprint\]](#) Reflective Frame Selection Agent for Long-Video Understanding

Focus Areas: Long-Video Understanding; Reflective Agents; Tool-Augmented Retrieval

- Developed RACER, a training-free reflective agentic framework that decomposes long-video frame selection into query interpretation by a lightweight Vid-LLM and evidence localization through an embedding-based retrieval tool.
- Improved performance by 4.3–6.6 points over uniform sampling across Video-MME, MLVU, and LongVideoBench with four Vid-LLMs, with gains up to 11.6 points on MLVU under a 32-frame answering budget.

[\[ICLR'26\]](#) Mitigating Hallucinations in Multimodal LLMs

Focus Areas: Hallucination Mitigation; Visual Encoder Bias; Inference-Time Intervention

- Identified encoder-side causes of hallucinations and proposed SHIELD, an inference-time intervention combining token reweighting/subtraction with adversarial contrastive decoding, achieving a 27% relative reduction in captioning hallucinations across multiple MLLM families.

[\[EMNLP'25\]](#) Training-Free Adaptation of Image-Pretrained MLLMs for Video Understanding

Focus Areas: Efficient Video Understanding; Token Compression; Question Decomposition

- Developed D-CoDe, a plug-and-play framework that adapts image-pretrained MLLMs for VideoQA through question decomposition and dynamic token/frame compression, achieving a 29% relative improvement in egocentric VideoQA accuracy with LLaVA-NeXT-7B.

INDUSTRY EXPERIENCE

Adobe Research | Vision-Language Lab | Research Intern | San Jose, CA
Mentors: Dr. [Zhaowen Wang](#), Dr. [Shraman Pramanick](#) May 2026 – Nov 2026 (Expected)

[\[Ongoing\]](#) Reflective Multimodal Agents for Visual Evaluation and Refinement

Focus Areas: Multimodal Agents; Reinforcement Learning; Reflection and Visual Refinement

- Identified limitations in holistic visual composition understanding and reading-order reasoning in existing VLMs and agents; constructed 1.7K training pairs, 9K+ perturbation pairs, and a 200-pair benchmark.
- Post-trained a Qwen3.5-9B multimodal reflection agent with GRPO and LoRA, enabling hierarchical reasoning and refinement through selective editing and render-based verification.
- Achieved 84.3% mean pairwise accuracy on Crello and internal Adobe product data, outperforming Gemini-3.6-Flash (73.0%), Claude-Opus-4.8 (66.9%), and GPT-5.6-Sol (59.0%).

Adobe Research | Vision-Language Lab | Research Intern | San Jose, CA
Mentors: Dr. [Zhaowen Wang](#), Dr. [Simon Jenni](#), Dr. [Jing Shi](#) May 2025 – Nov 2025

[\[ECCV'26\]](#) [\[Patent Filed\]](#) Compositional Layout Understanding in MLLMs

Focus Areas: Vision-Language Models; Compositional Reasoning; Structured Visual Understanding

- Identified semantic drift and structural ambiguity as key challenges in compositional visual reasoning, and developed MASON with metadata-visual grounding and 3D structural perception, achieving 26% and 15% relative improvements over Gemini-2.5-Pro and GPT-5/GPT-o3, respectively, using Qwen2.5-VL-7B as the backbone.
- Constructed CoDeLayout, a ~20K-sample benchmark, via an automated pipeline for compositional structure mining and multimodal QA generation.

PUBLICATIONS

First-Author Papers

- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Jianglin Lu, Qihua Dong, Hailing Wang, Huimin Zeng, Mingyuan Zhang, Yun Fu. “RACER: Reflective Agent Coupling Query Interpretation and Tool-Based Retrieval for Frame Selection in Long Video Understanding.” *Preprint*. [\[Paper\]](#)
- **Yiyang Huang**, Zhaowen Wang, Simon Jenni, Jing Shi, Yitian Zhang, Yizhou Wang, Yun Fu. “Beyond Atomic Layouts: Compositional Design Understanding with Vision-Language Models.” **ECCV 2026**. [\[Paper\]](#) [\[Code\]](#) [\[Project Page\]](#)
- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Mingyuan Zhang, Liang Shi, Huimin Zeng, Yun Fu. “Distorted or Fabricated? A Survey on Hallucination in Video LLMs.” **ACL 2026** (Findings). [\[Paper\]](#) [\[Repo\]](#)
- **Yiyang Huang**, Liang Shi, Yitian Zhang, Yi Xu, Yun Fu. “SHIELD: Suppressing Hallucinations in LVLM Encoders via Bias and Vulnerability Defense.” **ICLR 2026**. [\[Paper\]](#) [\[Code\]](#)
- **Yiyang Huang**, Yizhou Wang, Yun Fu. “D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression and Question Decomposition.” **EMNLP 2025** (Main). [\[Paper\]](#) [\[Code\]](#)
- **Yiyang Huang**, Xuefeng Liang, Chaowei Fang. “CALLip: Lipreading using Contrastive and Attribute Learning.” **ACM MM 2021**. [\[Paper\]](#)

Selected Co-Authored Papers

- Hailing Wang, Jianglin Lu, Xu Ma, **Yiyang Huang**, Yun Fu. “StreamVLM: Adapting Offline Vision-Language Models for Streaming Understanding.” *Under review*.
- Mingyuan Zhang, Yue Bai, Yifan Wang, **Yiyang Huang**, Yun Fu. “Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models.” *Preprint*. [\[Paper\]](#)
- Shuai Zou, Xuefeng Liang, **Yiyang Huang**. “LipReading for Low-resource Languages by Language Dynamic LoRA.” **ICASSP 2025**. [\[Paper\]](#)

ACADEMIC SERVICE

Conference Reviewing: ICLR; NeurIPS; ACL; EMNLP; IEEE Face & Gesture (FG)

Journal Reviewing: ACM TKDD

TEACHING

Teaching Assistant

DS 5110: Essentials of Data Science	Fall 2025
DS 5020: Fundamentals of Linear Algebra and Probability	Spring 2026
DS 5020: Fundamentals of Linear Algebra and Probability	Fall 2026

SKILLS

Languages: Python, $\text{\LaTeX}$

ML Frameworks: PyTorch, Hugging Face Transformers, Hugging Face Datasets

Training/Fine-Tuning Libraries: Accelerate, PEFT, bitsandbytes

RL/Post-Training Frameworks: TRL, verl, OpenRLHF

Inference & Evaluation: vLLM, Imms-eval

Video Processing: ffmpeg, PyAV, decord

Systems/Infrastructure: Linux, Git, Slurm

AWARDS

National Scholarship, China	2021
Outstanding Student, Xidian University	2022
1st Prize, Undergraduate Computer Design Competition (National Level), China	2021
2nd Prize, RoboMaster National Robotics Competition, China	2019
3rd Prize, ICRA AI Challenge	2019